Thermodynamics and thermal energy

Jens Oluf Jensen

Department of Energy Conversion and Storage
Fysikvej, building 310
Technical University of Denmark
2800 Lyngby
Denmark
jojen@dtu.dk

Outline

- A brief introduction to thermodynamics
- The thermodynamic functions
- Carnot’s law
- Heat engines and heat pumps
- Change in thermodynamic functions of chemical reactions
- Energy contents of fuels
- Geothermal energy
- Ocean thermal energy conversion

Thermodynamics

What is thermodynamics?

In brief: physics of the dynamics of *heat* and *work*.

Key features

- Thermodynamics distinguish between *ordered* and *disordered* energy
- The degree of disorder is quantified by *entropy*.
- Entropy can be generated but never destroyed

In this context relevant for

- power generation from thermal energy sources.
- the function and efficiency of heat pumps.
- quantification of the energy content of a fuel available for a given application.

Capital letter or small cap symbols?

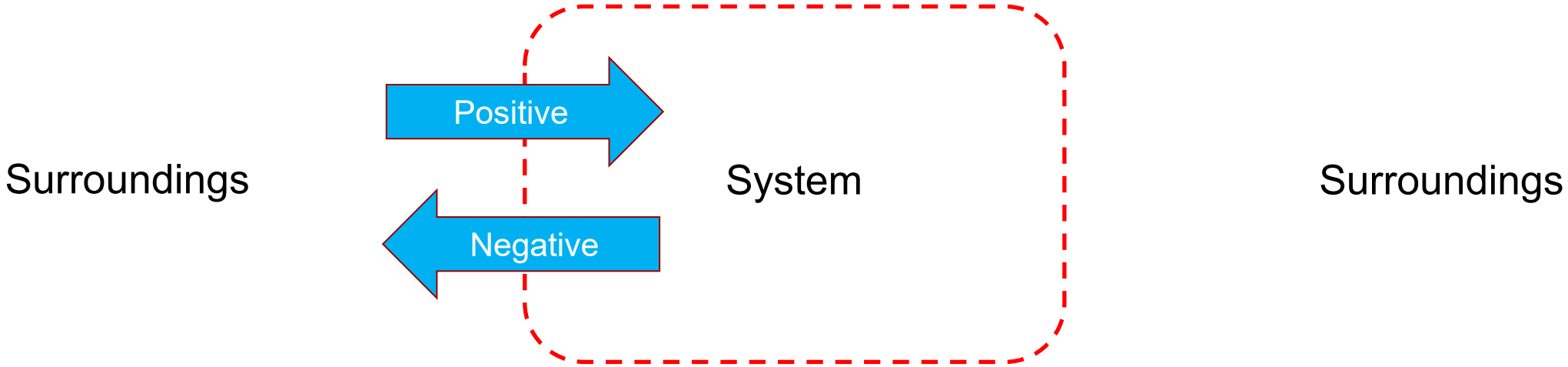
Difference between textbooks

Often *capital letter symbols* are used for absolute quantities and *small cap symbols* for either molar or mass specific quantities:

U: Internal energy (J)
u: Molar internal energy (J mol^{-1})
or mass specific internal energy (J kg^{-1})

C_V: Constant volume heat capacity (J K^{-1})
c_V: Constant volume heat capacity (J mol^{-1} K^{-1} or J kg^{-1} K^{-1})

The thermodynamic system



All is from the system's perspective

- Energy flows in are positive
- Energy flows out are negative

Conversion between macroscopic energy forms

Conversion from a to b: possible or not?

	Macroscopic, ordered			Disordered
b \ a	**Kinetic**	**Potential**	**Electrical**	**Thermal**
Kinetic		Yes	Yes	Partly
Potential	Yes		Yes	Partly
Electrical	Yes	Yes		Partly
Thermal	Yes	Yes	Yes	

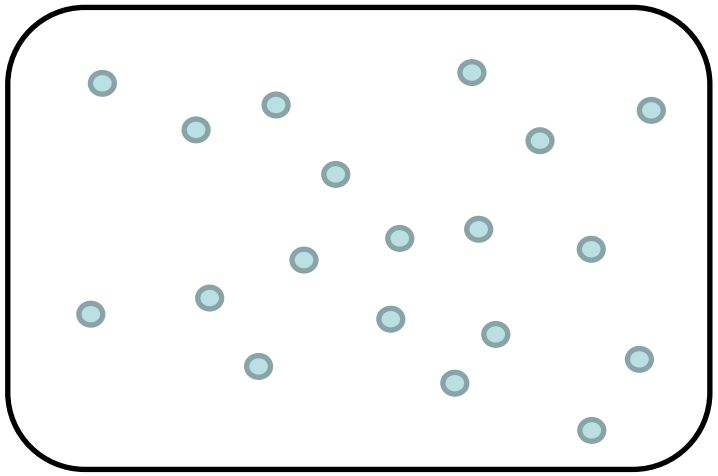
This is where thermodynamics is relevant

The most important thermodynamic functions

The most important thermodynamic functions

- Internal energy, U
- Enthalpy, H
- Entropy, S
- Gibbs free energy, G

Internal energy, *U*, and enthalpy, *H*



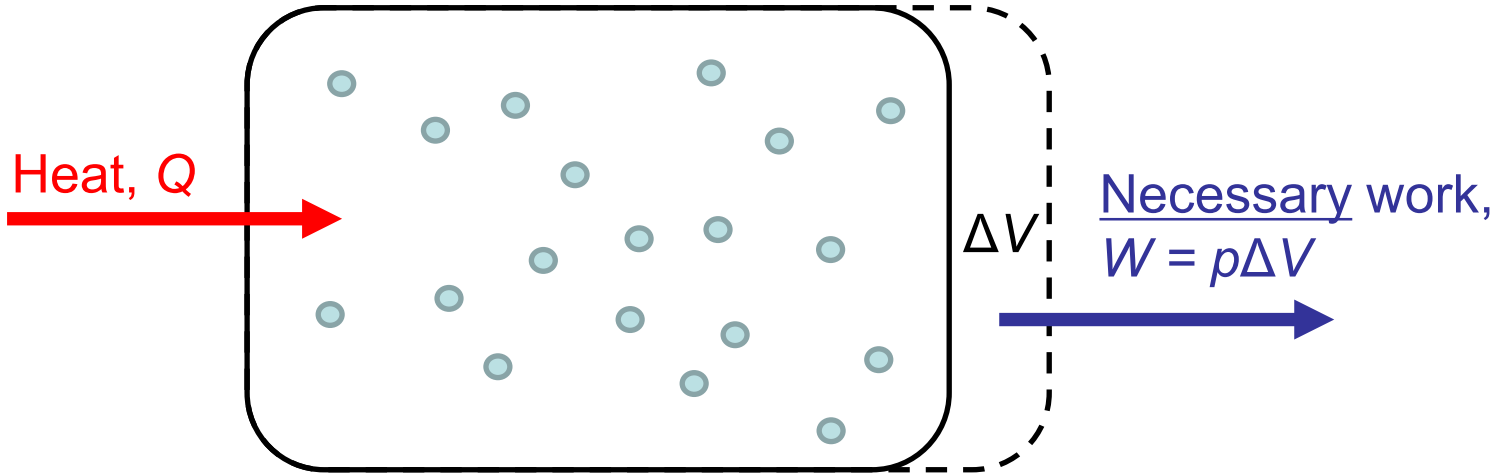
In principle all energy of the system

But in most cases, only the thermal energy is considered (kinetic and vibrational energy of the molecules).

U = the sum of the thermal energy of all atoms/molecules in the system.

H = *U* + the product of pressure and volume, $H \equiv U + pV$ (by definition)

The smart thing about enthalpy



Systems at constant pressure, *p*, change volume if energy is added/extracted.

This causes a change of volume (to keep the pressure), which changes the thermal energy.

Increase of energy ⇒ expansion ⇒ loss of part of the gained energy.

The lost energy equals the pressure-volume work on the surroundings ($p\Delta V$)

The pressure-volume term of the enthalpy increases by the same ($p\Delta V$)

Thus, enthalpy is changing like internal energy but unaffected by the necessary pressure-volume work.

By using enthalpy, we need not consider the necessary volume changes

Internal energy, *U* (same for *H*)

Absolute values of *U* and *H* are difficult to establish.

Which energies are relevant to include?

Not a problem. **We are normally only interested in changes, Δ*U* or d*U* (or Δ*H* or d*H*)**

U and *H* changes if energy is added to or removed from the system

- Heat is transferred to or from the system (*accelerates/decelerates molecular motion*)
- Work is done *on* or *by* the system (*accelerates/decelerates molecular motion*)

State functions

Functions (properties) that are depending on the *state of the system* only

The state is given by composition, pressure and temperature.

At any fixed state, the state function is fixed too.

Internal energy, *U*, and enthalpy, *H*, are state functions

No history: the system contains no information on how it was established

Why is this important?

Because calculation of a change from stage 1 to stage 2 can be done independently of how the change is made.

E.g., we can calculate the temperature change and the pressure change individually

Heat capacity, *C*

When a system is heated, how much does the temperature increase?

The temperature increase pr. energy or normally the energy change per temperature is given by the heat capacity, C

$$\Delta(\text{thermal energy}) = C\Delta T$$

Assuming for simplicity that C is constant

If the thermal energy in focus is the internal energy, then:

$$\Delta U = C_{\mathrm{v}}\Delta T$$

If the thermal energy in focus is the enthalpy, then:

$$\Delta H = C_{\mathrm{p}}\Delta T$$

More correctly: $\Delta H \approx C_{\mathrm{p}}\Delta T$

$\mathrm{d}H = C_{\mathrm{p}}\mathrm{d}T$ is exact.

Then: $\Delta H = \int_{T_1}^{T_2} C_{\mathrm{p}}\mathrm{d}T$

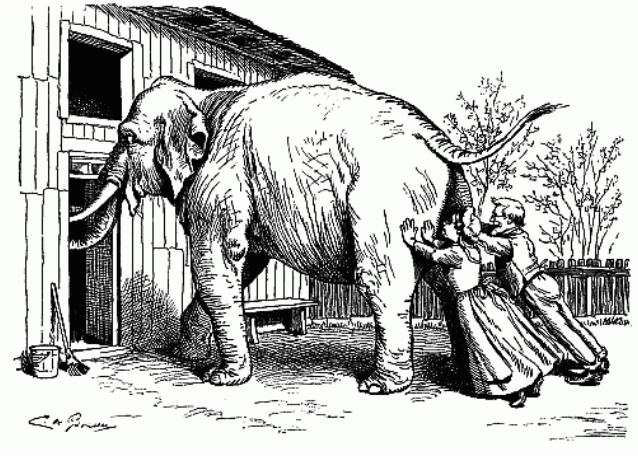
where C_{v} is the heat capacity measures at constant volume

And C_{p} is the heat capacity measures at constant volume

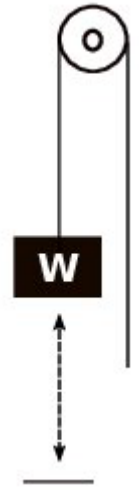
What is work?

To perform work is to create ordered **movement against an opposing force**







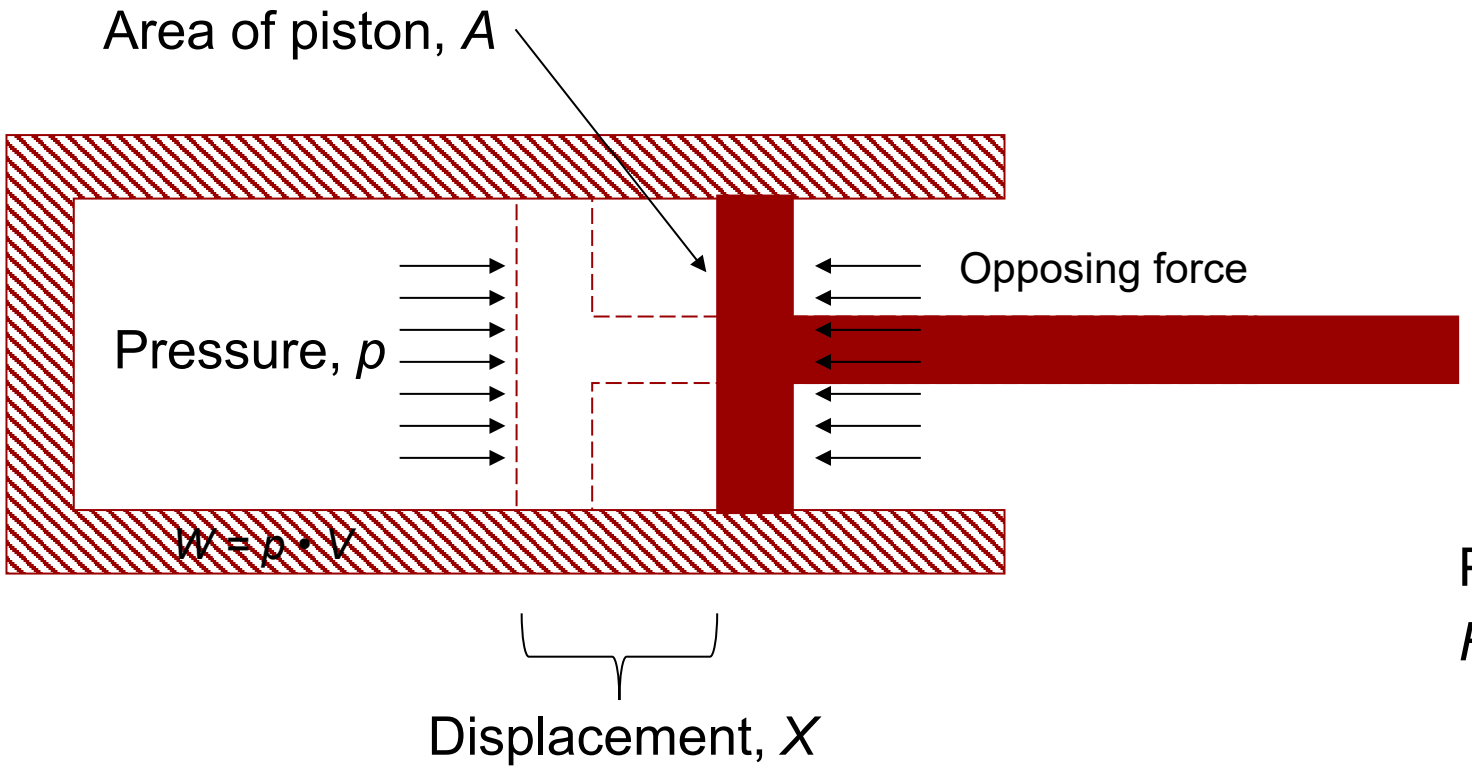


The dimension is energy

Work = force times displacement

A process is performing work if it can (in principle) lift a weight

Pressure-volume work



Work: $W = F \bullet X$

Pressure = Force / area = F / A

$F = p \bullet A$

$W = F \bullet X = p \bullet A \bullet X = p \bullet V$

$$W = p \bullet V$$

Heat and work

In thermodynamics, heat and work are *energy in transition*

The system can receive heat and work from the surroundings

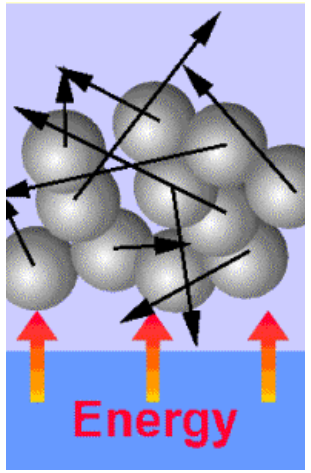
This changes the internal energy or enthalpy

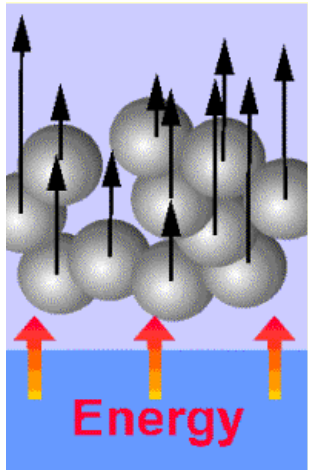
The system does *not contain* heat nor work (*not state functions*)

Heat

Disordered thermal energy transferred to/from the system.

Molecules are accelerated randomly





Work

Ordered mechanical (electrical) energy transferred to/from the system.

Molecules are accelerated ordered

Entropy, *S*

- Entropy is a measure of how much energy and matter are *spread* or *scattered*.
- Often referred to as *disorder*

Thermal energy is scattered over the atoms and molecules which contain it

Higher temperature: more energy to scatter, higher entropy

Atoms and molecules, in particular gasses are spread over a given volume

Larger volume: more spreading, more entropy

Entropy is a state function

$$S_{gas} >>> S_{liquid} > S_{solid}$$

Reversibility

The term "*reversibility*" is generated form the idea that some processes have *ability to reverse*.

In thermodynamics, a reversible process is one that can equally easy go forward and backward.

Reversible processes

- Happens through an equilibrium situation
- Have no preferred direction (no net driving force)
- Generate no entropy
- Are infinitively slow
- Can be reversed by an infinitesimal change
- Are thermodynamic abstractions

Irreversible processes

- Proceed in one direction ("downhill")
- Generate entropy
- Proceed at a finite (perhaps fast) speed
- Cannot be reversed by an infinitesimal change
- Are real

See Appendix on reversibility in DTU Learn

The 1. law of thermodynamics

Energy is conserved

Energy cannot be created nor destroyed

No indication whether a process is possible or not

The 2. law of thermodynamics

Entropy increases

The **entropy** of an isolated system increases for *irreversible* processes and remains constant for *reversible* processes.

The entropy of isolated systems, **never** decreases

Gibbs free energy, *G*

The thermodynamic function, Gibbs free energy, *G*, finds many uses in chemistry.
Again, we are only interested in changes, ΔG.

In this context, we only use the Gibbs free energy as maximum work one can extract from a chemical process under ideal conditions.

The concept behind

1. If a process generates energy (enthalpy) it is not always possible to convert all the enthalpy to work.
2. If the process leads to lower entropy after all the generated energy is removed, then it cannot happen.
3. Consequently, Some of the generated energy must be used for generating the missing entropy.

Gibbs free energy stats how much energy is available for work after the entropy balance is maintained.

Gibbs free energy change, ΔG

Process: combustion of hydrogen: $2H_2(g) + O_2(g) \rightleftarrows 2H_2O(l)$

Energy released: Δh_r = enthalpy change of the reaction (here molar, i.e., per mole)

If used for heating:

Δh_r can be fully utilized. No entropy problems. **Maximum heat = Δh_r**

If used for work:

Entropy must be considered

The process has net reduction in the number of gas molecules. This is a reduction in entropy.

This missing entropy must be (re)generated by "wasting" some energy as heat generating thermal entropy. The rest of the energy can be converted into work.

If the missing entropy of the reaction is Δs_r, then the necessary heat to be dissipated is $T\Delta s_r$,

$$\textbf{Maximum work} = \Delta g_r = \Delta h_r - T\Delta s_r$$

Conversion between heat and work

The heat engine – Carnot's law

The principles behind Carnot's law

Thermal energy contains entropy

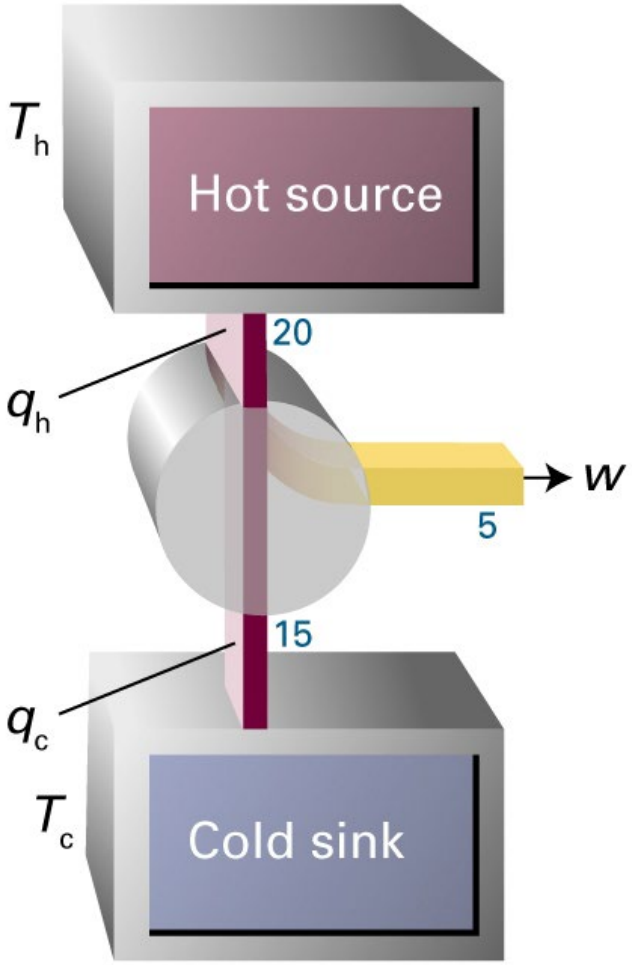
Work does not contain entropy

Complete conversion of thermal energy into work is destruction of entropy

Thus, part of the heat must be used to generate the missing entropy

Carnot's law tells us how much of the heat we can convert into work

(Requirement: satisfaction of the 2nd law)



$$\text{Efficiency}, \eta = \frac{\text{What you get}}{\text{What you pay}}$$

$$\text{Efficiency}, \eta = \frac{\text{Work}}{\text{High temp. heat}}$$

Carnot's Law

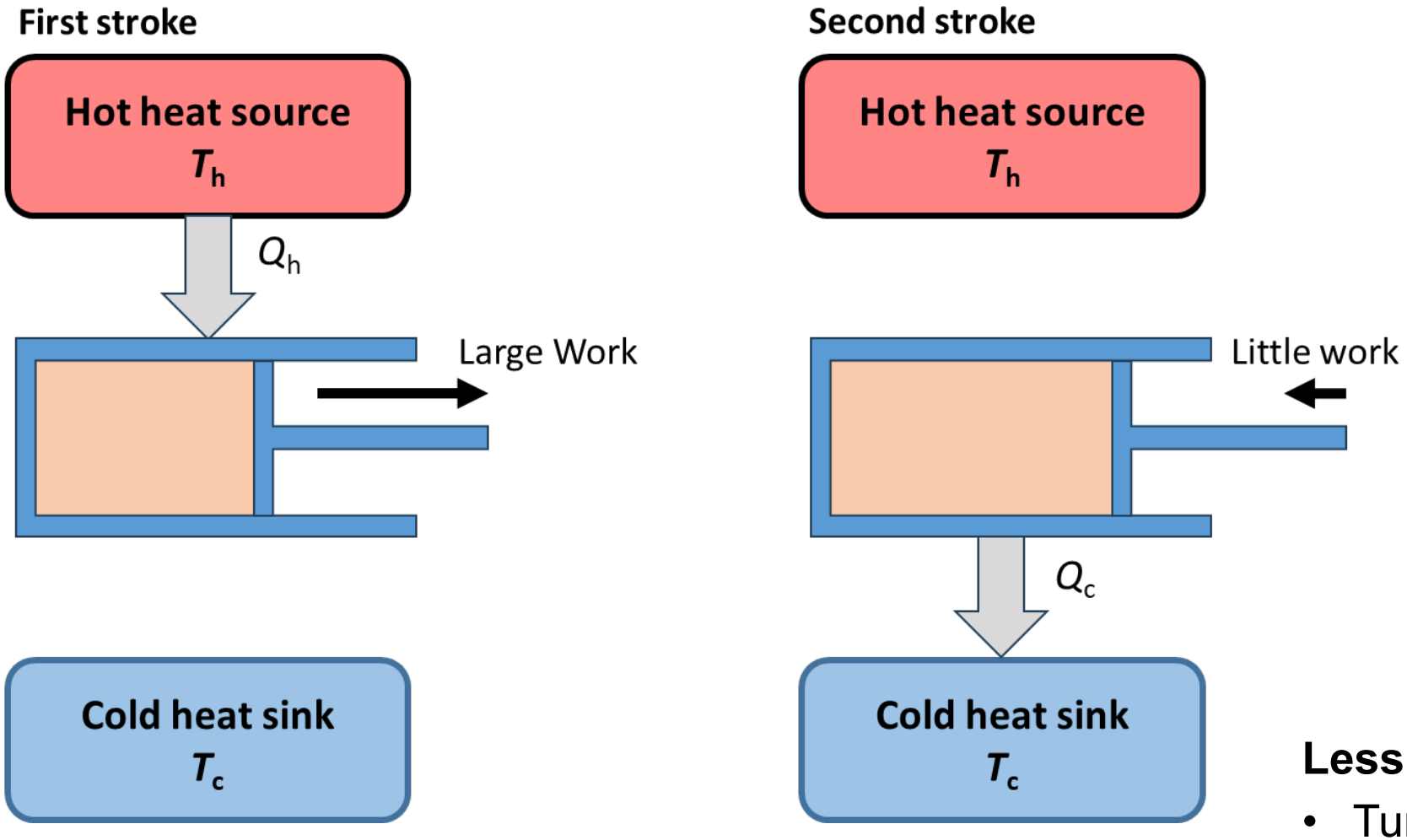
$$\eta_{\text{carnot}} = \frac{T_h - T_c}{T_h} = 1 - \frac{T_c}{T_h}$$

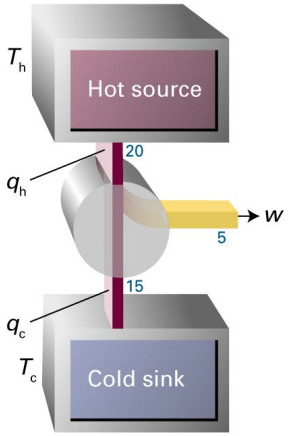
$$W_{\max} = Q_h \eta_{\text{carnot}}$$

(No indication of how to construct such a heat engine)

The heat engine – Carnot's law

$$W_{\max} = Q_h \eta_{\text{carnot}}$$





Less intuitive heat engines are

- Turbines
- Internal combustion engines

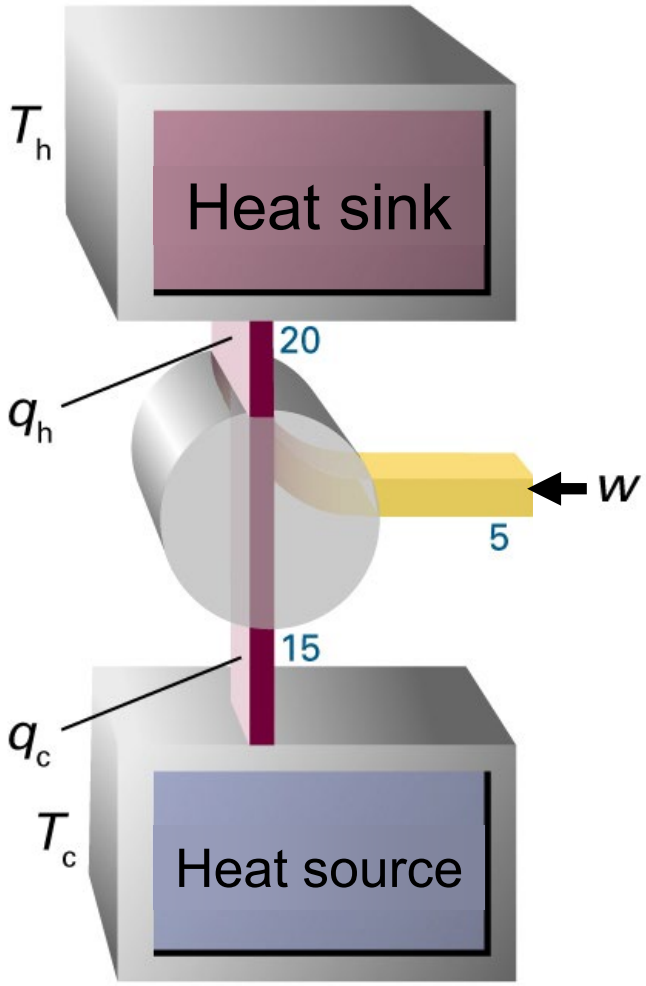
The principle of a heat pump

A heat pump increases the temperature of low-grade heat for heating purposes.

The opposite process of the heat engine.

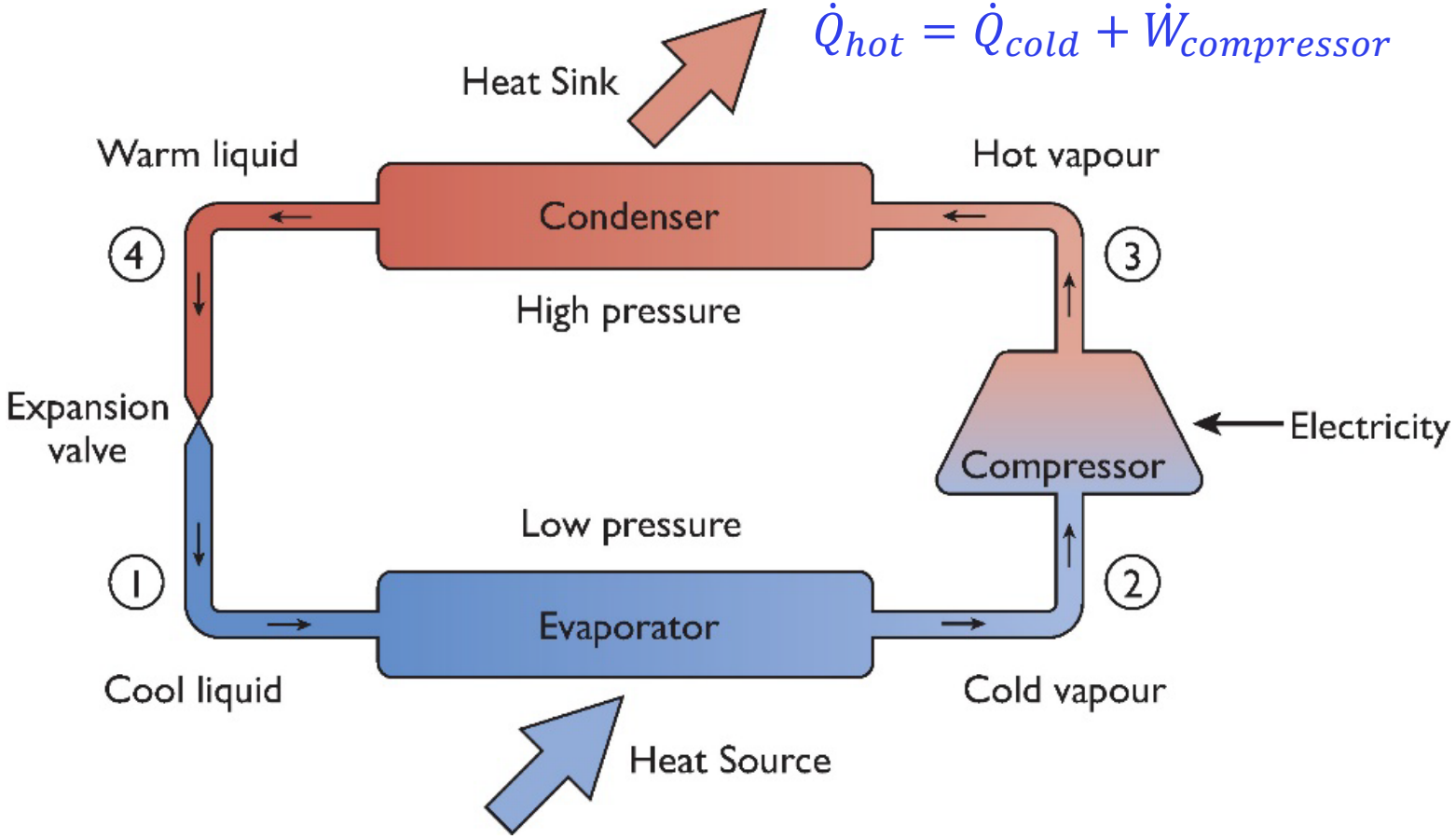
The heat flow is now from cold to hot.

This is not spontaneous and must be forced by addition of work.



The principle of a heat pump

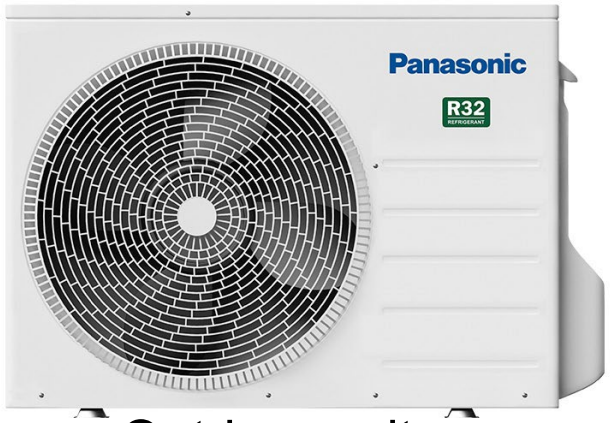
$$\dot{Q}_{hot} = \dot{Q}_{cold} + \dot{W}_{compressor}$$

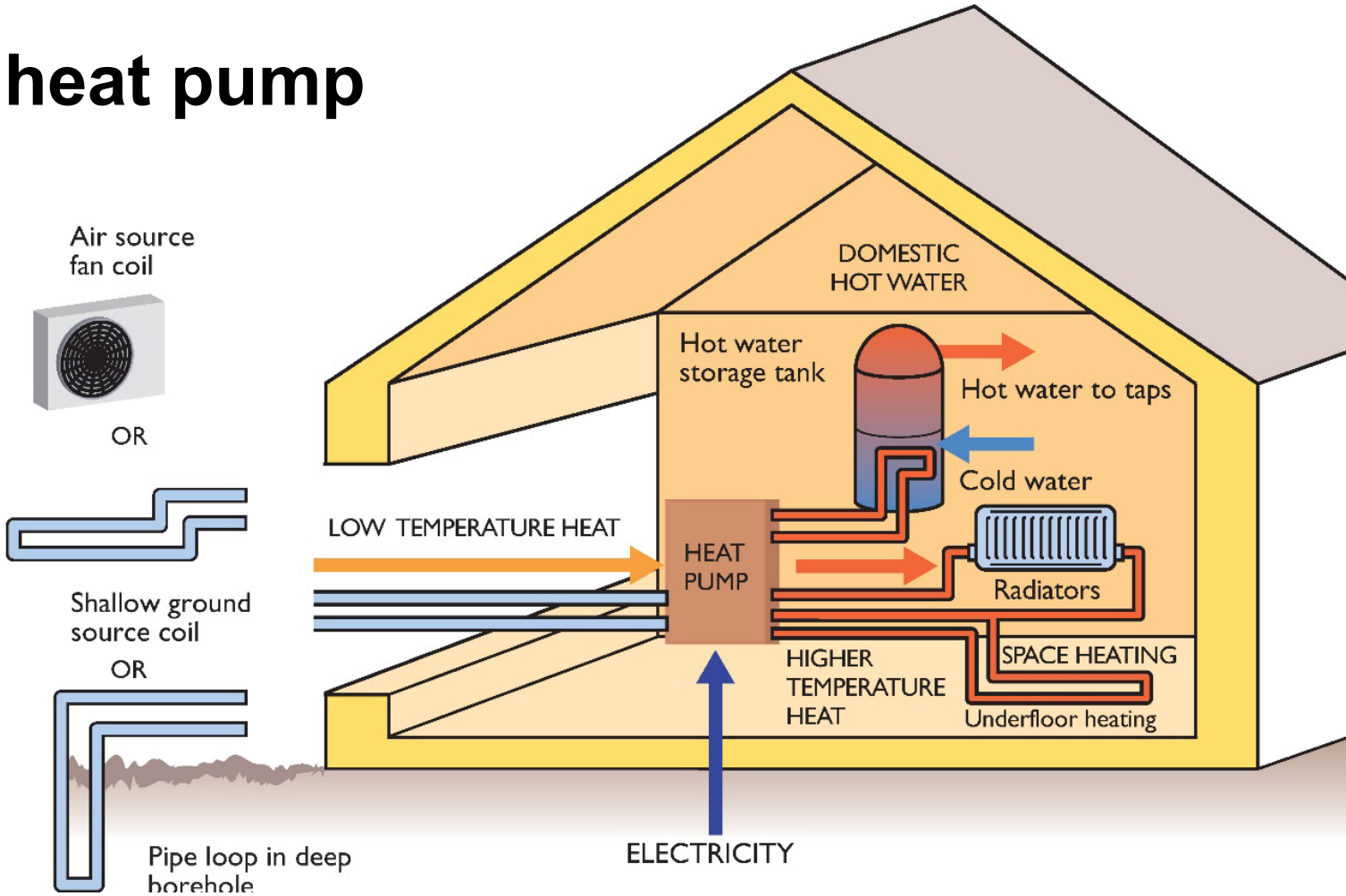


Low temperature heat upgraded and used at higher temperature

The principle of a heat pump







Efficiency: Coefficient of Performance (COP) > 1

$$\mathrm{COP} = \frac{\dot{Q}_{\mathrm{hot}}}{\dot{W}_{\mathrm{compressor}}} = \frac{P_{\mathrm{hot}}}{P_{\mathrm{electrical}}}$$

$P =$ power

Change in thermodynamic functions of chemical reactions Δh_r, Δs_r and Δg_r

Balancing chemical reactions

Criteria:

Same number of each atom on each side (mass balance)

Unbalanced: $CH_4 + O_2 \rightleftarrows CO_2 + H_2O$

Balanced: $CH_4 + \underline{2}O_2 \rightleftarrows CO_2 + \underline{2}H_2O$

Unbalanced: $N_2 + H_2 \rightleftarrows NH_3$

$N_2 + H_2 \rightleftarrows \underline{2}NH_3$

Balanced: $N_2 + \underline{3}H_2 \rightleftarrows 2NH_3$

Unbalanced: $C_2H_5OH + O_2 \rightleftarrows CO_2 + H_2O$

$C_2H_5OH + O_2 \rightleftarrows \underline{2}CO_2 + H_2O$

$C_2H_5OH + O_2 \rightleftarrows 2CO_2 + \underline{3}H_2O$

Balanced: $C_2H_5OH + \underline{3}O_2 \rightleftarrows 2CO_2 + 3H_2O$

See Appendix on Learn

Δh_r, Δs_r and Δg_r of chemical reactions

Chemical reaction, e.g., $A + B \rightarrow C + D$

Example for Δh_r (same for Δs_r or Δg_r)

1. All species assigned a value of enthalpy, h, (molar quantities from a table)
2. Δh_r is the sum of the enthalpies of all *formed species* (products, C and D) minus the enthalpies of all consumed species (reactants, A and B)
3. All enthalpy values of species must be multiplied with the stoichiometric coefficients, ν.

General form: $\nu_A A + \nu_B B + \cdots \rightarrow \nu_M M + \nu_N N + \cdots$ $\quad \nu_i = \text{stoiciometric coefficients}$

$\Delta h_r = \nu_M h(M) + \nu_N h(N) + \ldots - \nu_A h(A) - \nu_B h(B) - \ldots$

$\Delta s_r = \nu_M s(M) + \nu_N s(N) + \ldots - \nu_A s(A) - \nu_B s(B) - \ldots$

$\Delta g_r = \nu_M g(M) + \nu_N g(N) + \ldots - \nu_A g(A) - \nu_B g(B) - \ldots$

Sum of products minus sum of reactants

$$\Delta x_r = \sum_{\text{prod.,i}} \nu_i \, x_i \; - \sum_{\text{react.,j}} \nu_j \, x_j$$

Example

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$,

$\Delta h_r = h(CO_2) + 2h(H_2O) - h(CH_4) - 2h(O_2)$ $\Delta h_r = -890$ kJ/mol

General: $$\Delta h_r = \sum_{\text{Prod.}} \nu_i \Delta h_i - \sum_{\text{React.}} \nu_i \Delta h_i$$

Δh_r = Molar enthalpy of the reaction

Standard enthalpy of formation

Problem: How to assign enthalpy values to the species when we only measure changes?

Solution: Standard enthalpy of formation

The *standard enthalpy of formation,* Δh_f, of a substance is defined as:

The enthalpy change of a chemical process forming the species from its constituting elements in their most stable state at p = 1 bar and T = 298.15 K (= 25 °C).

Example, ethanol: element 1 + element 2 + element 3 ⇄ Molecule of interest

$$\Delta h_f(CH_3CH_2OH) = \Delta h_r\ (2C + ½O_2 + 3H_2 \rightleftarrows CH_3CH_2OH)$$

Standard enthalpy of formation

Example, ethanol: element 1 + element 2 + element 3 ⇄ Molecule of interest

$$\Delta h_f(CH_3CH_2OH) = \Delta h_r\ (2C + \tfrac{1}{2}O_2 + 3H_2 \rightleftarrows CH_3CH_2OH)$$

How does that help?
Do we just shift the problem to other species?

$$\Delta h_f(C) = \Delta h_r\ (C \rightleftarrows C) = 0$$

$$\Delta h_f(O_2) = \Delta h_r\ (O_2 \rightleftarrows O_2) = 0$$

$$\Delta h_f(H_2) = \Delta h_r\ (H_2 \rightleftarrows H_2) = 0$$

(By definition: formation from the elements)

Standard enthalpy of formation of any pure element is zero

Gibbs free energy of reaction

Works the same way as enthalpy of reaction:

$$\Delta g_r^0 = \sum_{\text{prod.,i}} \nu_i \, \Delta g_{f,i}^0 - \sum_{\text{react.,j}} \nu_j \, \Delta g_{f,j}^0$$

Standard Gibbs free energy of formation is defined just like standard enthalpy of formation

Note:

Gibbs free energy of reaction also determines if the process can happen spontaneously or not.

1. If Δg_r is negative, then the process can happen and even produce work. The 2. law is respected.
2. If Δg_r is positive, then the $T\Delta s_r$ is larger than Δh_r. Then the total energy liberated, Δh_r, is insufficient to satisfy the second law (even if all of it is wasted) and the process cannot happen unless forced by addition of work.

Entropy of reaction

Entropy change of reaction works like enthalpy and free energy:

$$\Delta s_r^0 = \sum_{\text{prod.}} \nu_i \, \Delta s_i^0 - \sum_{\text{react.}} \nu_i \, \Delta s_i^0$$

However, there is no Standard entropy of formation

Instead, *third law entropies* are used for all species.

These area absolute entropy values calculated from the process of bringing the species from absolute zero temperature to 298.15 K

These values are also tabulated

Thermodynamic table

Note:

- Not all values are known
- For pure elements, ΔG_f and ΔH_f are zero
- Some units are kJ/mol, othes J/mol K
- The relation $\Delta G = \Delta H - T\Delta S$ will not fit as relation between the values ΔG, ΔH and S in the table because ΔG and ΔH are values of formation and S are third law entropies defined differently, and they are not ”deltas”.

Standard Thermodynamic Properties of Chemical Substances

5-7

Molecular formula	Name	Crystal $\Delta_f H°$ kJ/mol	Crystal $\Delta_f G°$ kJ/mol	Crystal $S°$ J/mol K	Crystal C_p J/mol K	Liquid $\Delta_f H°$ kJ/mol	Liquid $\Delta_f G°$ kJ/mol	Liquid $S°$ J/mol K	Liquid C_p J/mol K	Gas $\Delta_f H°$ kJ/mol	Gas $\Delta_f G°$ kJ/mol	Gas $S°$ J/mol K	Gas C_p J/mol K
BrCs	Cesium bromide	-405.8	-391.4	113.1	52.9								
BrCu	Copper(I) bromide	-104.6	-100.8	96.1	54.7								
BrF	Bromine fluoride									-93.8	-109.2	229.0	33.0
BrF_3	Bromine trifluoride					-300.8	-240.5	178.2	124.6	-255.6	-229.4	292.5	66.6
BrF_5	Bromine pentafluoride					-458.6	-351.8	225.1		-428.9	-350.6	320.2	99.6
BrGe	Germanium monobromide									235.6			37.1
$BrGeH_3$	Bromogermane											274.8	56.4
BrH	Hydrogen bromide									-36.3	-53.4	198.7	29.1
BrHSi	Bromosilylene									-464.4			
BrH_3Si	Bromosilane											262.4	52.8
BrH_4N	Ammonium bromide	-270.8	-175.2	113.0	96.0								
BrI	Iodine bromide									40.8	3.7	258.8	36.4

Note: capital symbols are use, but the values are molar

Energy content of a fuel

Heating value

Standard enthalpy of combustion, Δh_c^0

Fuel made from carbon, hydrogen and oxygen: $\text{Fuel} + x\text{O}_2 \rightarrow y\text{CO}_2 + z\text{H}_2\text{O} + \text{enthalpy}$

All carbon to CO_2, all hydrogen to H_2O, all oxygen to CO_2 or H_2O

$$\text{Fuel} + x\text{O}_2 \rightarrow y\text{CO}_2 + z\text{H}_2\text{O}(\text{l}) + \text{enthalpy}$$

$$\text{Fuel} + x\text{O}_2 \rightarrow y\text{CO}_2 + z\text{H}_2\text{O}(\text{g}) + \text{enthalpy}$$

The heating value of a fuel is the total enthalpy liberated ($-\Delta h_c^0$)

Higher heating value (HHV) when product water is liquid

Lower heating value (LHV) when product water is vapour

Enthalpy of combustion

Octane combustion: $C_8H_{18} + 12.5O_2 \rightarrow 8CO_2 + 9H_2O$

$$\Delta h^0_{c,C8H18} = \Delta h^0_{\mathrm{r}} = 8\Delta h^0_{\mathrm{f,CO2}} + 9\Delta h^0_{\mathrm{f,H2O(l)}} - \Delta h^0_{\mathrm{f,C8H18}} - 12.5\Delta h^0_{\mathrm{f,O2}}$$

With the table values:

$\Delta h^0_{\mathrm{f,CO2}} = -393.51\ \mathrm{kJ\ mol^{-1}}$

$\Delta h^0_{\mathrm{f,H2O(l)}} = -285.83\ \mathrm{kJ\ mol^{-1}}$,

$\Delta h^0_{\mathrm{f,C8H18}} = -249.9\ \mathrm{kJ\ mol^{-1}}$,

$\Delta h^0_{\mathrm{f,O2}} = 0.0\ \mathrm{kJ\ mol^{-1}}$,

the standard enthalpy of combustion, Δh^0_{c}, becomes -5479.65 kJ mol^{-1}.

Maximum heat and work of a fuel

The maximum heat obtainable from a fuel equals $-\Delta h_c^0$ of the combustion process

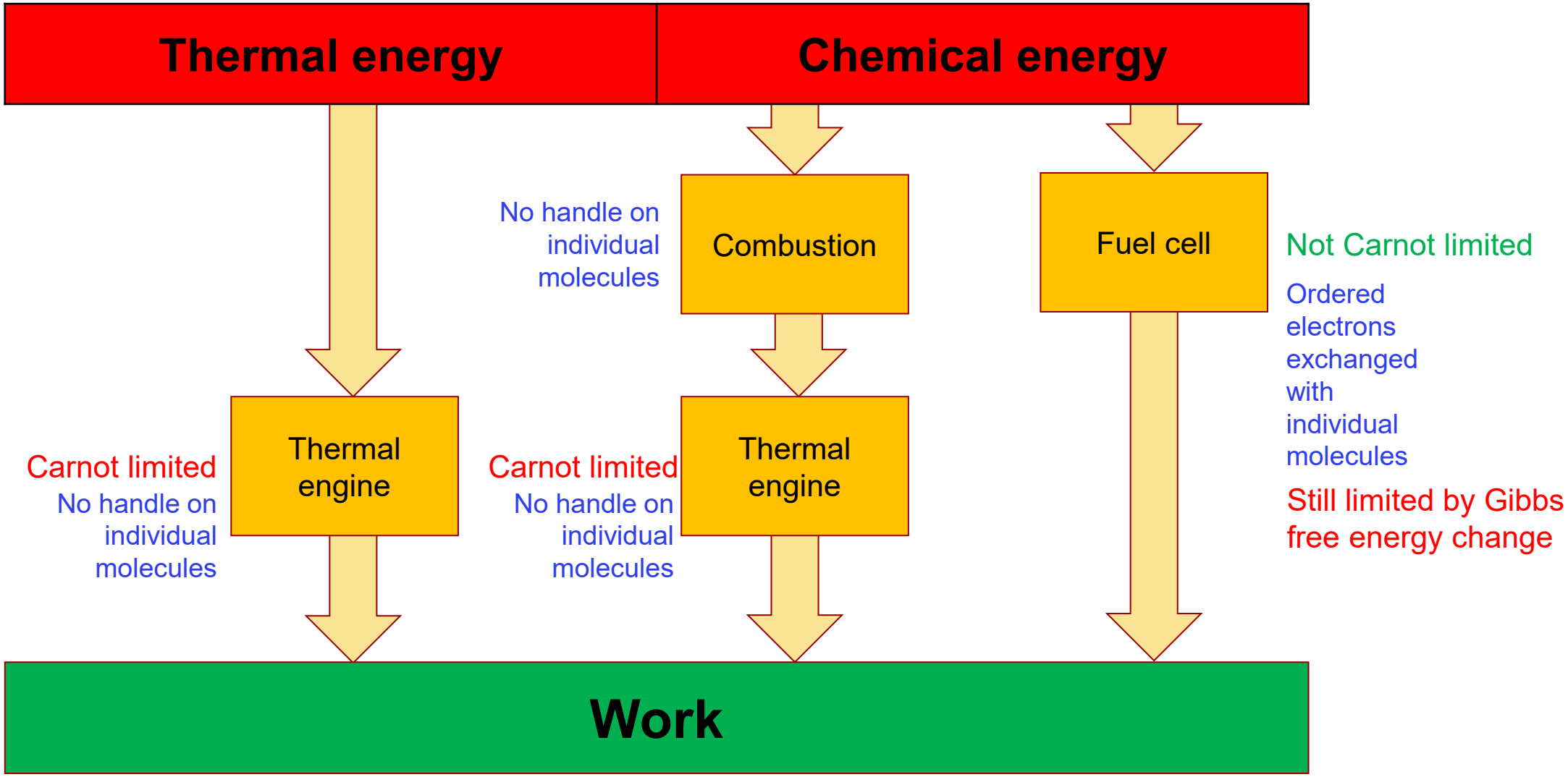
The maximum work obtainable from a fuel equals $-\Delta g_c^0$ of the combustion process

Octane combustion: $C_8H_{18} + xO_2 \rightarrow yCO_2 + zH_2O$

$C_8H_{18} + 12.5O_2 \rightarrow 8CO_2 + 9H_2O$

$$w_{max} = \Delta g^0_{c,C8H18} = \Delta g^0_{\mathrm{r}} = 8\Delta g^0_{\mathrm{f,CO2}} + 9\Delta g^0_{\mathrm{f,H2O(l)}} - \Delta g^0_{\mathrm{f,C8H18}} - 12.5\Delta g^0_{\mathrm{f,O2}}$$

Generation of work from thermal or chemical energy

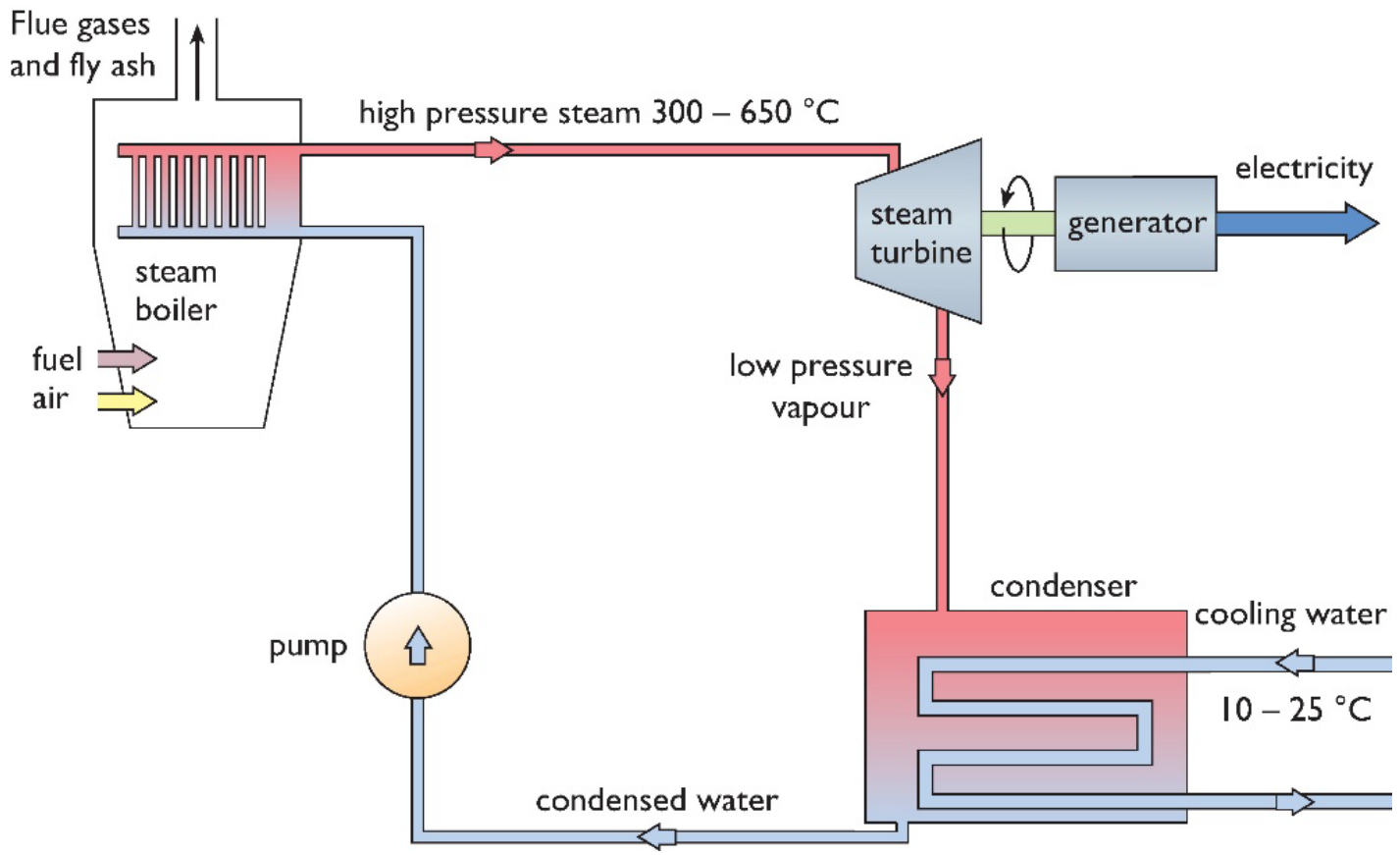


Summary of thermodynamics

- The change of **enthalpy** of a process is
  - the total energy released at constant pressure
  - the energy available for heating
- The **entropy** is the degree of disorder and the degree of unavailability for work
- The change of **(Gibbs) free energy** of a process is the maximum work obtainable
- The **signing** of (Gibbs) free energy determines if a process is spontaneous
- **Carnot's law** gives the maximum fraction of heat energy that can form work
- All **heat engines** and heat to work processes are limited by Carnot
- **Heat pumps** have over 100% efficiency called COP
- Thermodynamic function for chemical reactions are calculated from tabulated values

Thermal power plants
- in brief

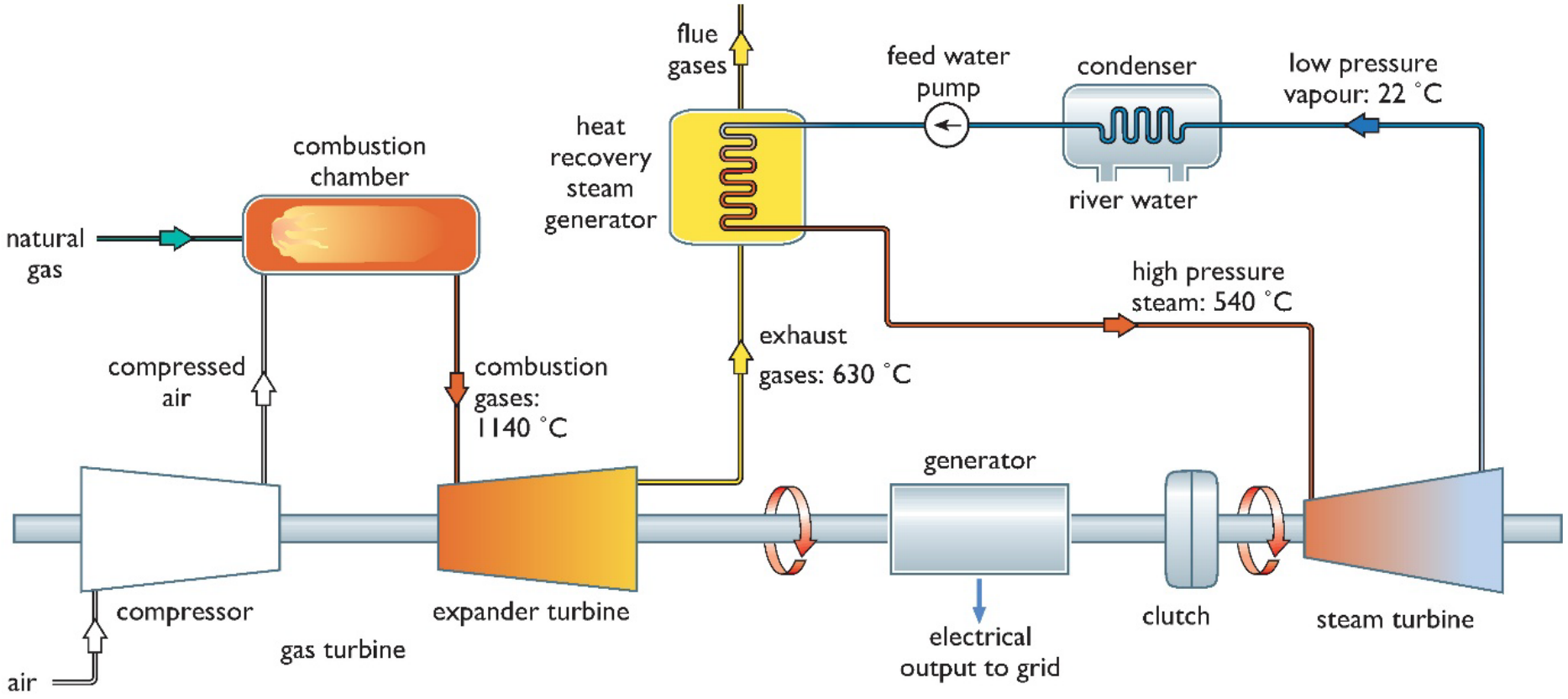
Thermal power plants, the Rankine cycle



Corrosion issues in contact with hot steam in the boiler limits the steam temperature to 650 °C

The moderate temperature limits the Carnot efficiency

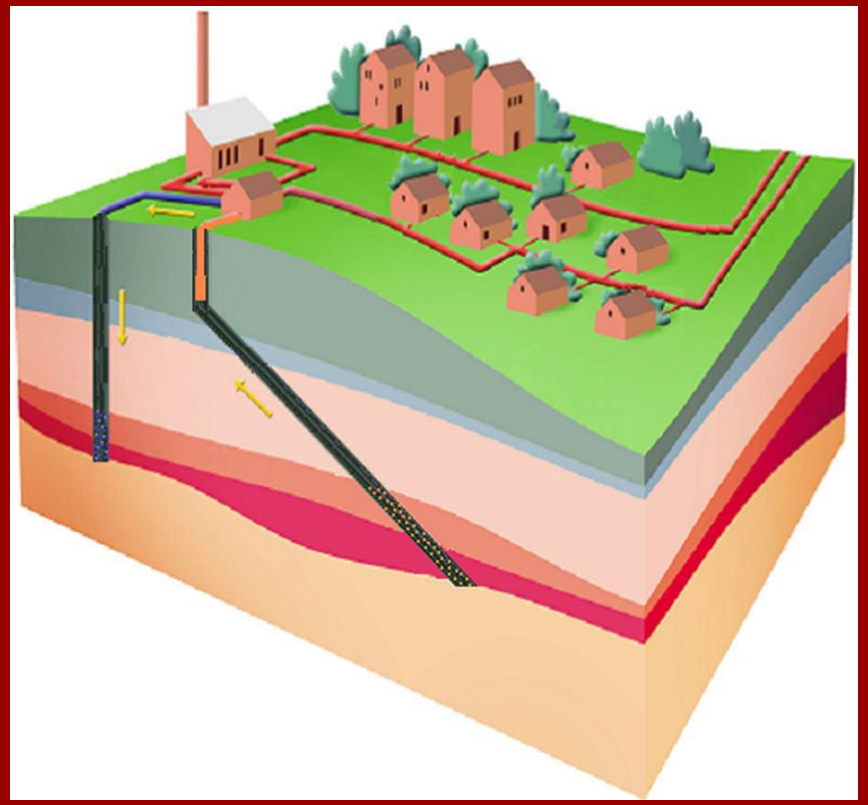
Thermal power plants, combined cycle



With a dry gas turbine, higher temperature can be reached.

This gives higher Carnot efficiency

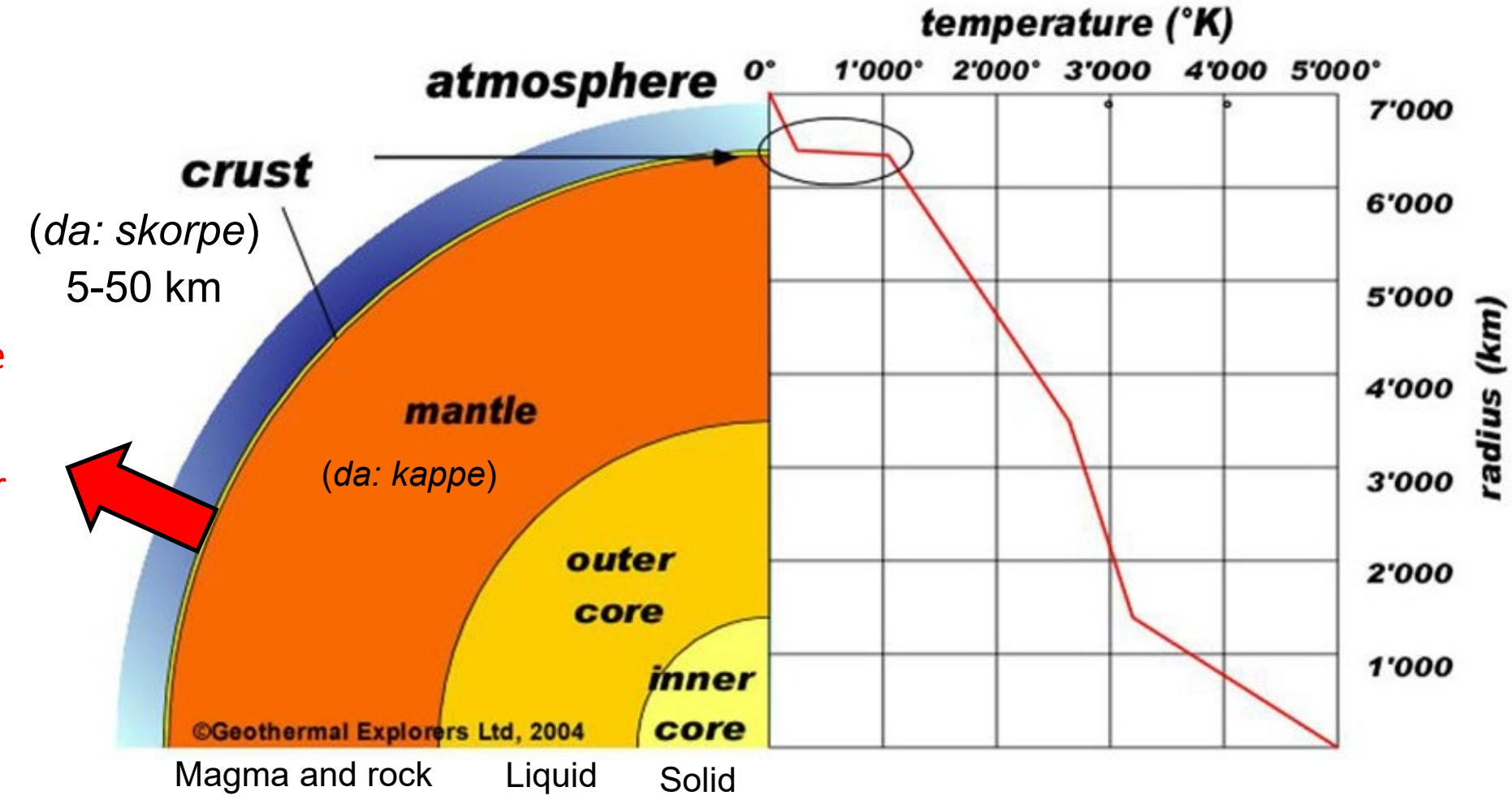
Geothermal energy



The temperature profile of planet Earth

Heat flow through the crust: 0.073 W m^{-1}

(in DK, 100 PJ per year or 2/3 of the district heating)



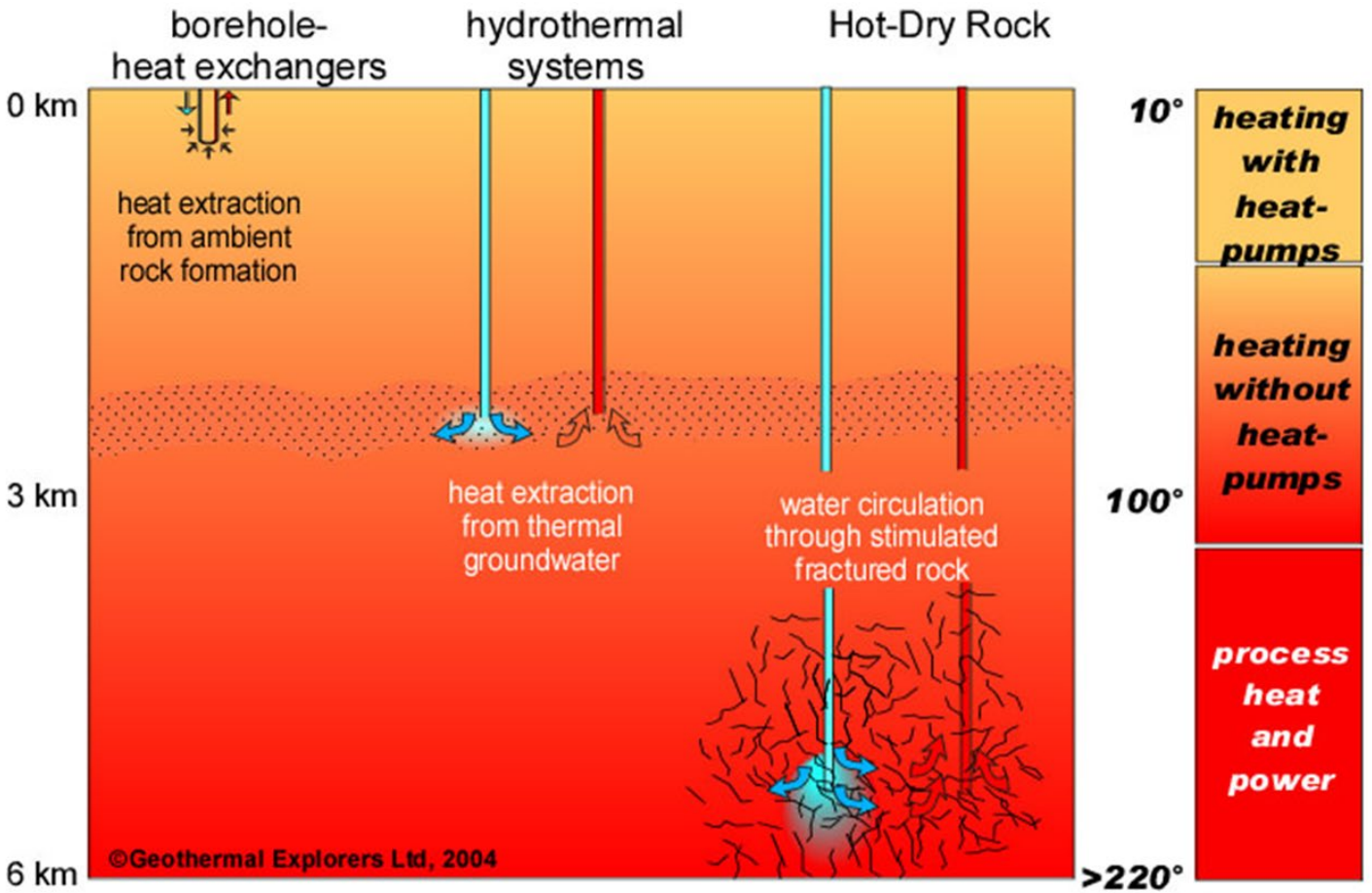
Three kinds of geothermal technology

1-3 m, 10-15 ºC.
Heating and cooling via heat pumps

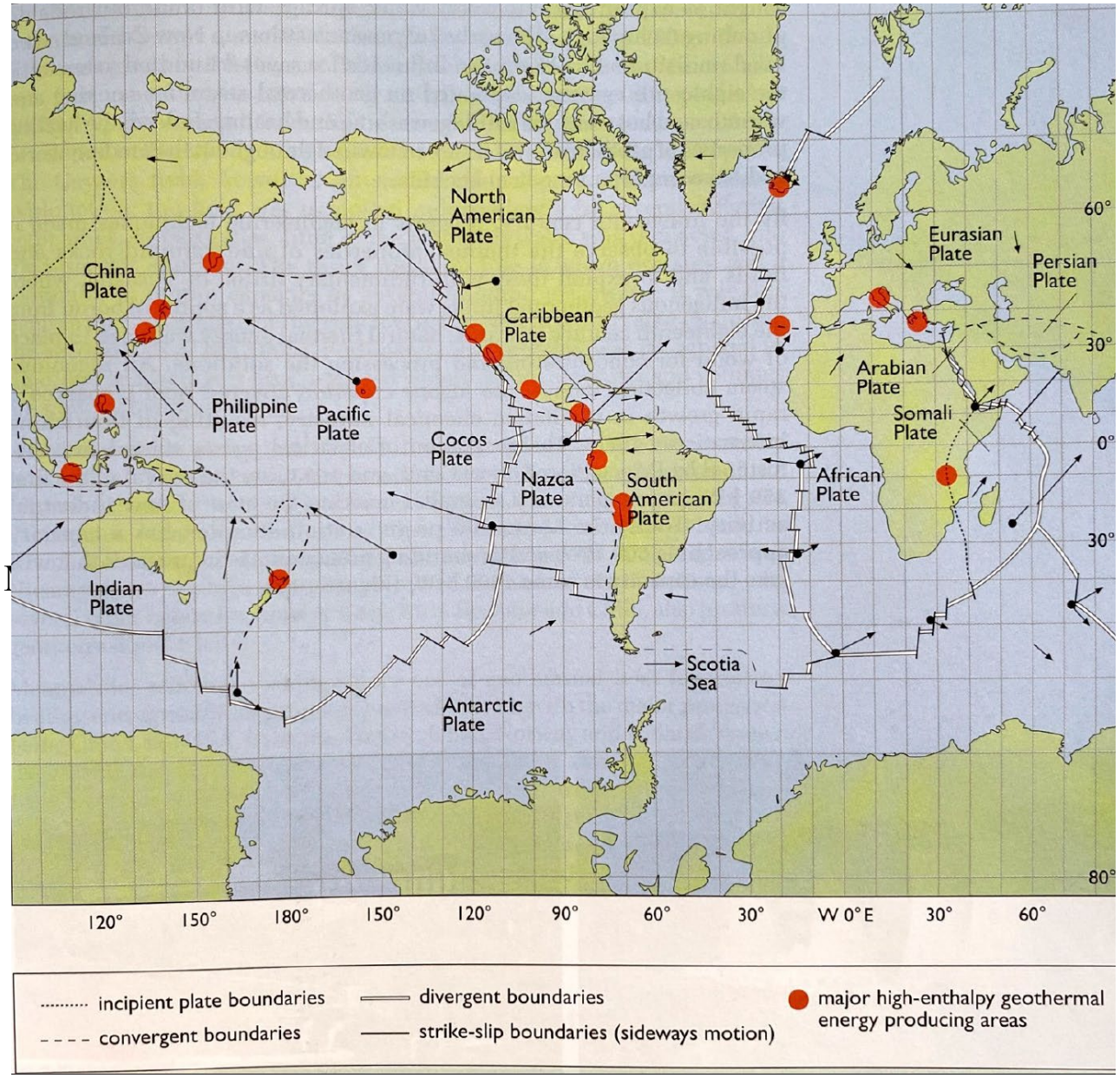
1-3 km, 30-100 ºC.
Hydrothermal heating

3-6 km, 150-350 ºC.
Power generation

Only 0-6 km is practical
Minimum thermal gradient is 25 ºC km^{-2}



Locations with accessible high temperature geothermal heat



James P. Warren. Deep geothermal energy. In Stephen Peake (ed), Renewable energy, 4ed. Oxford University Press 2018.

Geothermal power plants 2020

Ten nations having the most installed geothermal power Installed in 2020

1. U.S.A:	3,700 MW_e
2. Indonesia:	2,289 MW_e
3. Philippines:	1,918 MW_e
4. Turkey:	1,549 MW_e
5. Kenya:	1,193 MW_e
6. Mexico:	1,105 MW_e
7. New Zealand:	1,064 MW_e
8. Italy:	916 MW_e
9. Japan:	550 MW_e
10. Iceland:	755 MW_e

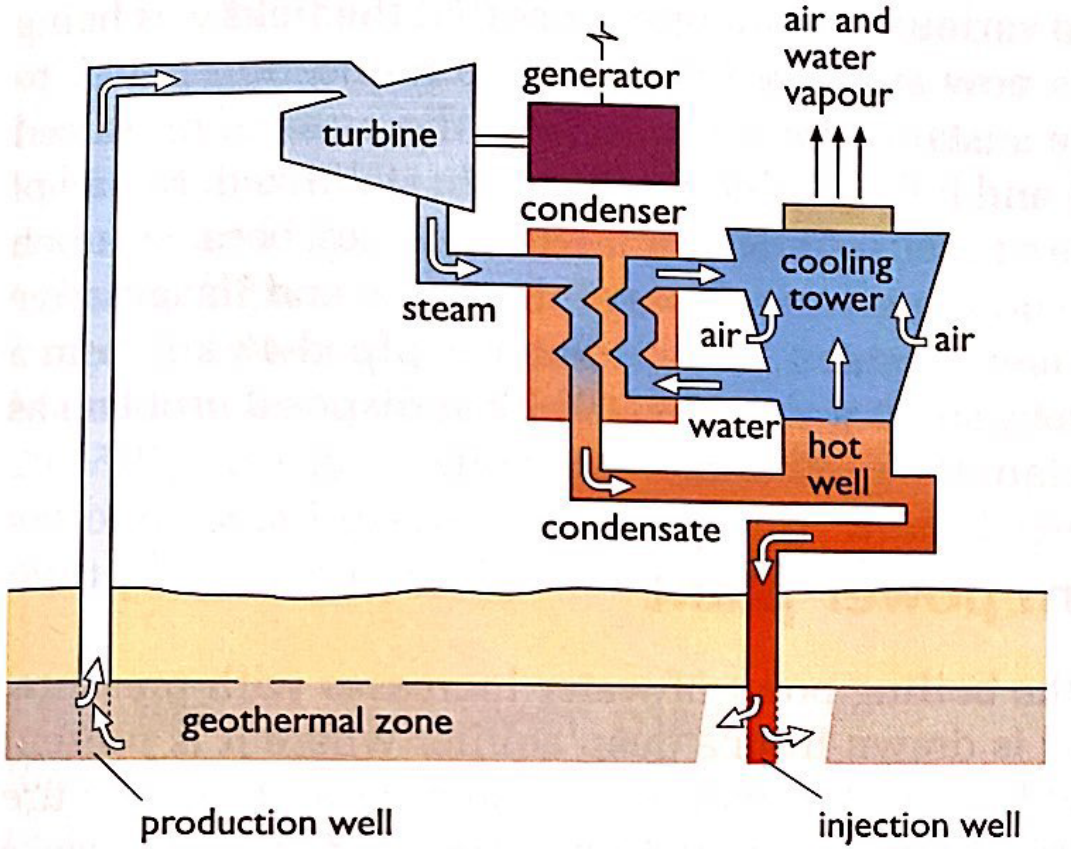


Gerald W. Huttrer, Proceedings World Geothermal Congress 2020, Reykjavik, Iceland, April 26 – May 2, 2020

Geothermal power generation

Dry steam power plant

- Heat source: steam with no water
- Typically, 180-350 ºC, 40-80 bar
- Direct powering of the turbine
- Condensate re-injected



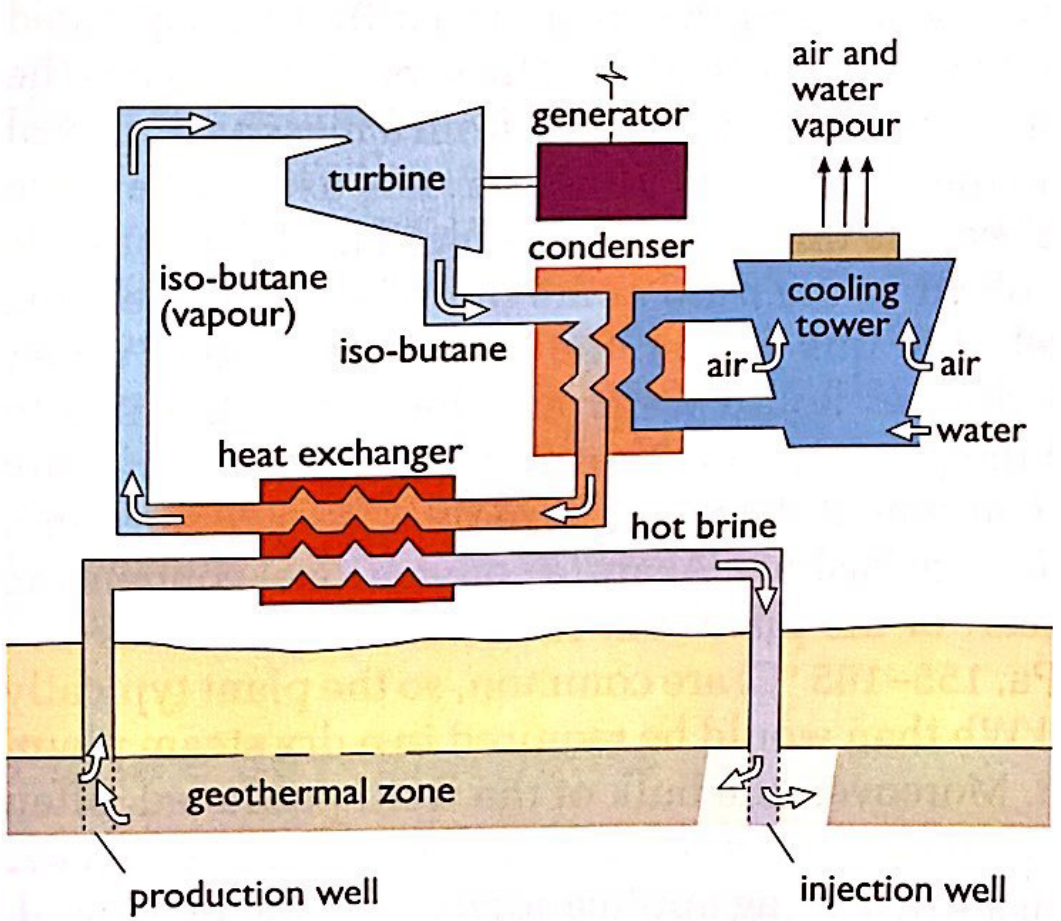
(a) dry steam power plant

James P. Warren. Deep geothermal energy. In Stephen Peake (ed), Renewable energy, 4ed. Oxford University Press 2018.

Geothermal power generation

Binary cycle / Organic Rankine Cycle power plant

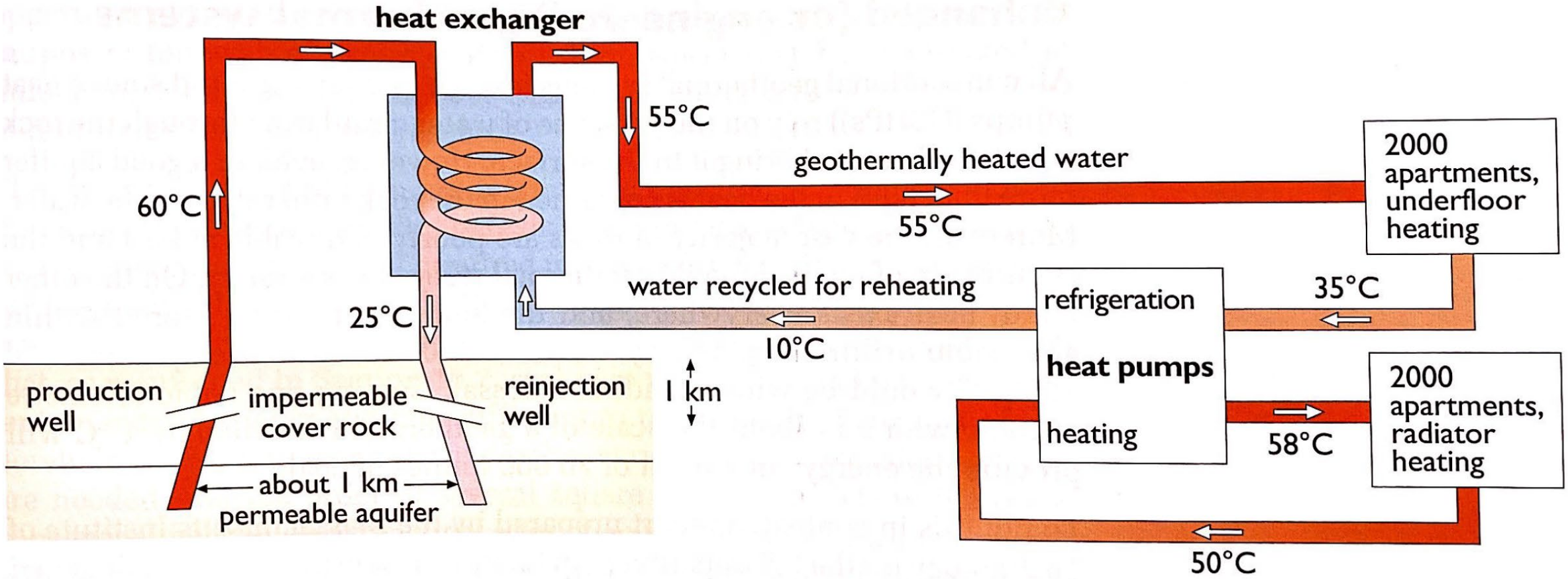
- Heat source: hot brine under pressure
- Heat exchange to a low boiling organic fluid (e.g. butane or pentane)
- Condensate re-injected



(c) binary cycle / organic rankine cycle power plant

James P. Warren. Deep geothermal energy. In Stephen Peake (ed), Renewable energy, 4ed. Oxford University Press 2018.

Low temperature systems for domestic heating



James P. Warren. Deep geothermal energy. In Stephen Peake (ed), Renewable energy, 4ed. Oxford University Press 2018.

Gerothermal heating in Denmark (only for heating)

Thisted, operated from 1984

Geothermal power: 7 MW (+ 10 MW from heat pump)

Depth: 1.25 km

Flow: 200 m^3 h^{-1}

Households: 2000

Temperature: 45 ºC



Sønderborg, operated from 2013

Geothermal power: 12.5 MW (+ MW from steam heat pump)

Depth: 1.2 km

Flow: m^3 h^{-1}

Households: 2400

Temperature: 48 ºC





Amager, operated from 2005

Geothermal power: 14 MW

(+ 13 MW from steam heat pump)

Depth: 2.6 km

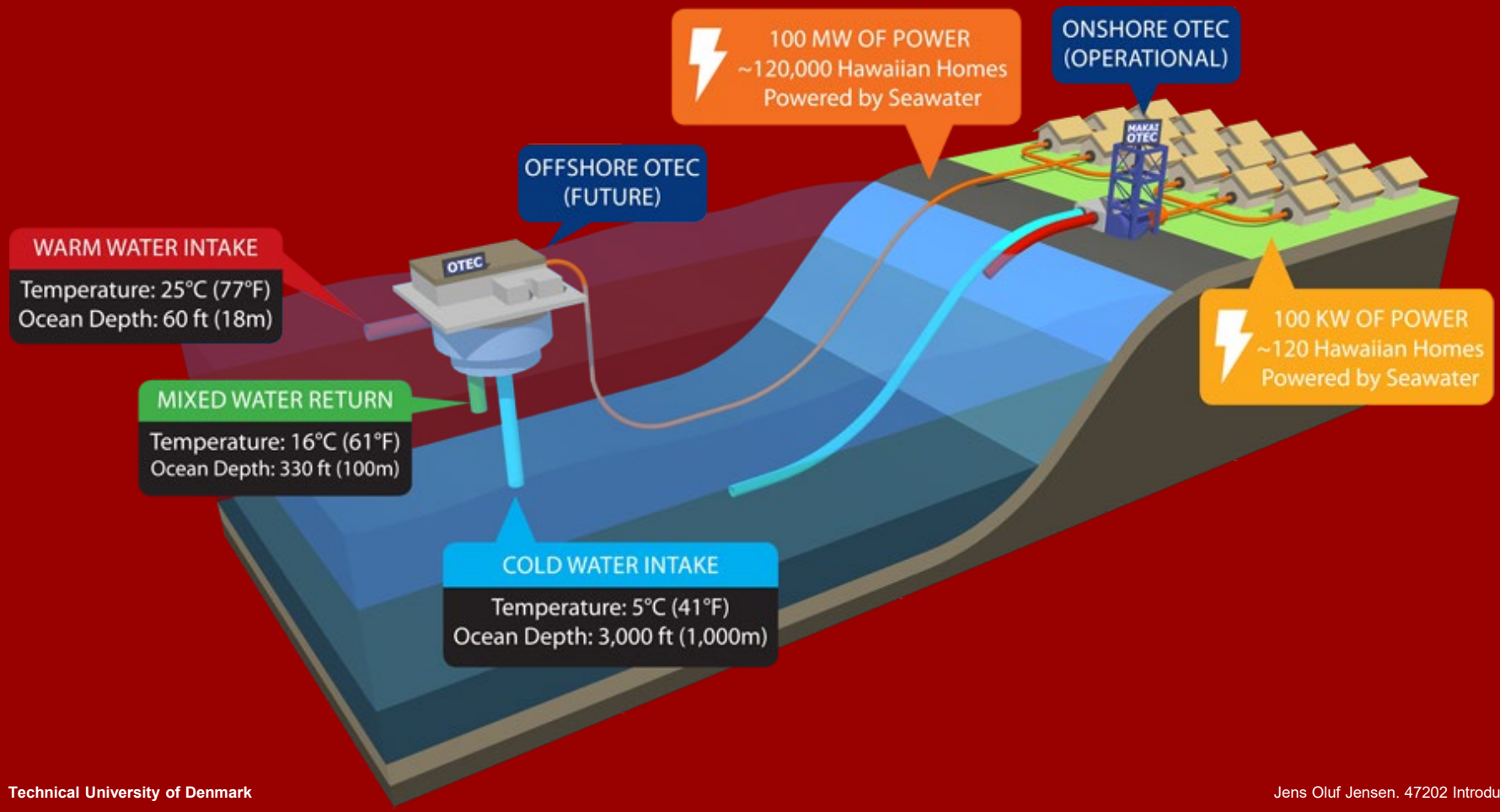
Flow: 230 m^3 h^{-1}

Households: 4600

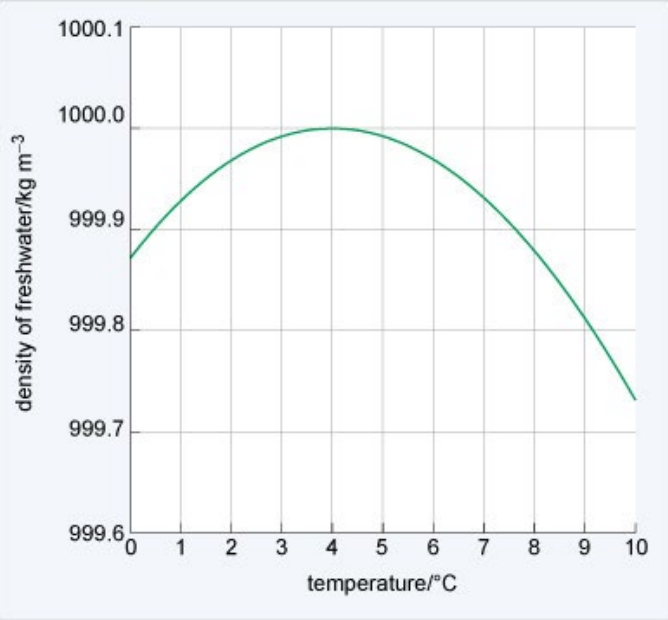
Temperature: 73 ºC

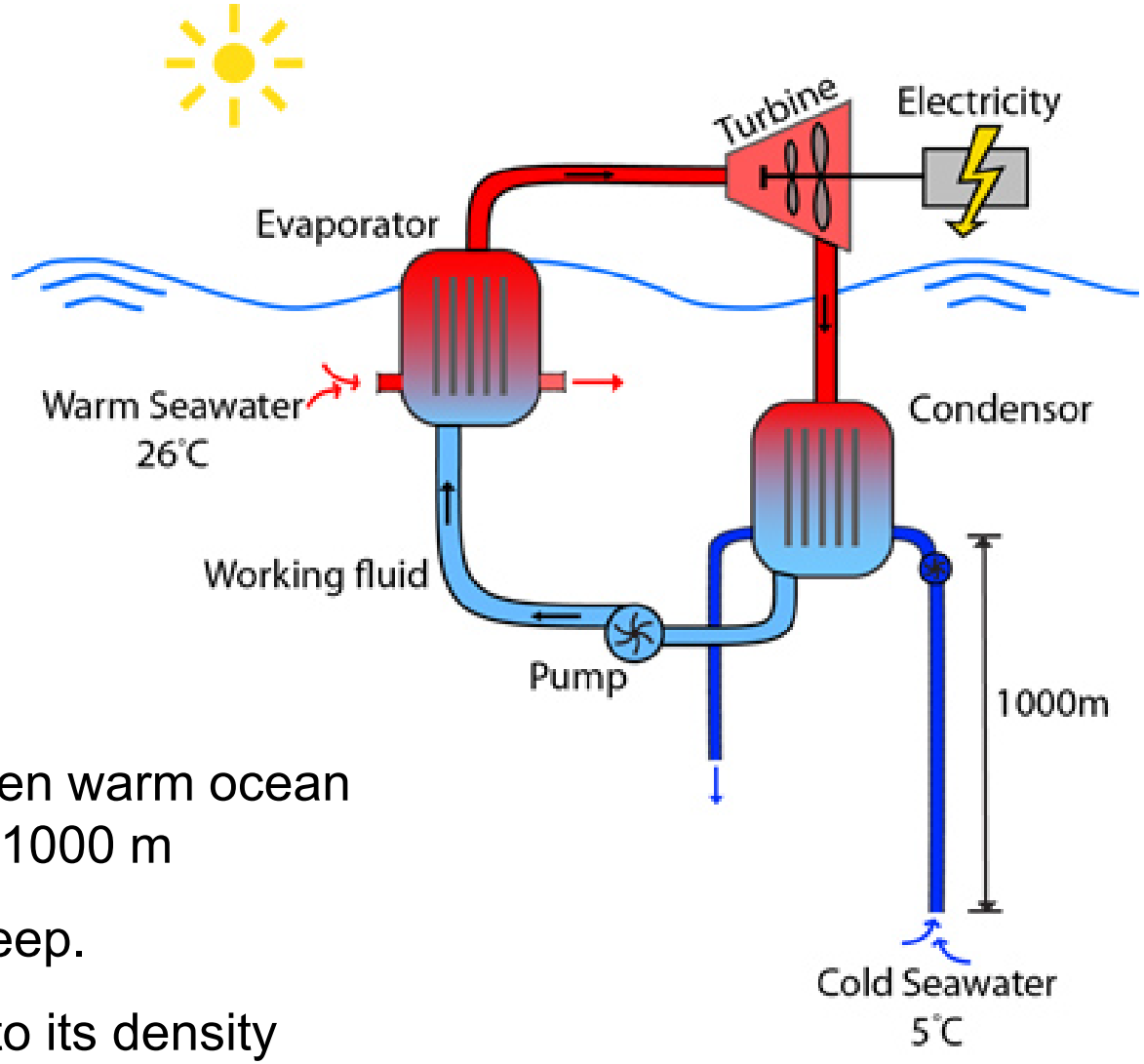


Ocean thermal energy conversion (OTEC)



The working principle of OTEC





Driving force: the temperature difference between warm ocean surface and the deep cold ocean of about 800–1000 m

Sunlight heats the surface layers, but not the deep.

Deep oceans contain water of around 4 C due to its density

Still in the research phase. Difficult to make economic. Very large water pumping rates